**Foundation of Blockchain Technology (IS501)**
**Pre-Reading Material – Lecture 36**

**Bitcoin Wallet**

**Estimated Reading Time:** 15–20 minutes

# Pre-Reading Material: Bitcoin Wallet

Estimated Reading Time: 5 minutes

## Introduction

A Bitcoin wallet is a software application or hardware device used to manage the keys required to send and receive Bitcoin. Unlike a physical wallet, it does not store Bitcoin itself. Bitcoin transactions and balances are recorded on the blockchain, while the wallet manages the keys that provide access to those funds.

## Components of a Bitcoin Wallet

- Private Key: A secret cryptographic key used to authorize Bitcoin transactions. It must be kept confidential.
- Public Key: A cryptographic key derived from the private key and used in verifying digital signatures.
- Bitcoin Address: An identifier shared with others to receive Bitcoin.
- Recovery Phrase: A sequence of words that can help restore access to a wallet if the original device is lost or damaged.

These components help users manage funds and authorize transactions securely.

## Types of Bitcoin Wallets

- Hot Wallet: Connected to the internet, such as a mobile or desktop wallet. It is convenient for everyday transactions but may face greater exposure to online threats.
- Cold Wallet: Keeps private keys offline, often using a hardware device. It reduces exposure to online attacks and is commonly used for long-term storage.


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Wallets may also be classified as software wallets, hardware wallets, or paper wallets, depending on how keys are managed and stored.

## Working of a Bitcoin Wallet

When a user wants to receive Bitcoin, the wallet provides a Bitcoin address to the sender. When sending Bitcoin, the wallet creates a transaction and uses the appropriate private key to generate a digital signature. The transaction is then broadcast to the network for verification and inclusion in the blockchain.

## Security and Importance

Bitcoin wallets provide convenient access to digital funds while supporting transaction authentication and ownership control. Users should protect their private keys and recovery phrases, maintain secure backups, and verify recipient addresses before sending transactions. Losing access to the required keys or recovery information may result in permanent loss of access to funds.

## Key Takeaway

A Bitcoin wallet manages the cryptographic keys needed to control Bitcoin recorded on the blockchain. Understanding wallet types, key management, and security practices is essential for using Bitcoin safely.

**Pre-Reading Questions**

1. What is a Bitcoin wallet?
2. Does a Bitcoin wallet store Bitcoin directly?
3. What is the role of a private key?
4. How does a hot wallet differ from a cold wallet?
5. Why is a recovery phrase important?


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